24/01/2026

# Final Year Project

## Youtube

Python for Bioinformatics - Drug discovery using machine learning and Data Analysis.

## Part-I:-

Data collection

Target Protein search
↓
chembl (Datasearch) through python database
↓
Bioactivity data ← Data preprocessing ← cleaned dataset

## Part-II:-

clean SMILES format (.csv file) → Calculate Lipinski in descriptors → Perform EDA (plots)

File #4 (2 class)
File #5 (3 class)
} Github

## Part-III:-

Descriptor calculation

read csv as data frame (File #4) → calculate descriptors → output (csv) export

## Part - IV

Model Building (csv as df) (File #6) → Remove Low Variance features → Data splitting → Build a Reg. Model ↓ scatterplot (Act. vs pred)

## Part - V :-

Model comparision | Build several regression models → make a Performance comparison plot

## Part - VI :

Deploy model | Web Application

Reg. model from Part - IV → Input molecule —smiles format→ code the web APP ⇌ save model to file

web APP ↓ Predicted Bioactivity

PART - I

## COLAB :-

File : CDD_ML_Part 1_bioactivity Data

1) Libraries = chembl_webresource_client (.new client)
Pandas as pd

2) Search for Target protein (SARS COV 2) (single Protein)

3) Select & retrieve bioactivity data for 3CLike Protease

4) bioactivity data save in CSV

5) Save in drive

Handle missing data

<u>Data pre-processing</u> :- Bioactivity data in $IC_{50}$

| Active | Inactive | Intermediate |
|---|---|---|
| <1000 nM | >10,000 nM | 1K - 10K nM |

Iterate molecule_chembl_id to a List
" " Canonical_smiles to a List
" " standard_value to List
bioactivity_class

↳ New dataframe by combining

Save CSV

bioactivity_preprocessed_data.csv

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Part-II

<u>EDA**</u>

1) Install conda
rdkit
upload bioactivity data

2) Calculate Lipinski descriptors (5)

3) Combine Dataframes (df + df_Lipinski)

4) Convert $IC_{50}$ → $pIC_{50}$

apply norm_value() for normalization

final $pIC_{50}$ values

Remove Intermediate class

EDA chemical space Analysis via Lipinski desc.

1) Import Lib (seaborn, matplotlib)
2) Freq. bioactivity_class plot
3) scatterplot of mw Vs. LogP
4) Boxplots for $pIC_{50}$ Vs. Bioactivity class
5) Statistical Analysis (Mann-whitney U-Test) (Active Vs. Inactive) for all descriptors i.e., LogP, mw, HA, HD, $pIC_{50}$

Results zip files

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Part-III (Descriptor Calculation)

Dataset prep

1) Download PaDeL (.zip, .sh) (Pandas)
2) Load bioactivity $pIC_{50}$ values (.csv)
3) Select canonical_smiles & molecule_chembl_id (save as molecule.smi)
4) Calculate fingerprint descriptors

cat padel.sh

bash padel.sh

(if take time download direct from github)

5) prepare X & Y data matrices

## Model Building

**Part-IV** Regression Random Forest

Import Lib.

Load dataset

Input features, output features

remove low variance features

Data split (80/20 ratio)

Build a Regression model using RF

scatter plot of Experimental vs. Predicted $IC_{50}$ values

## Model Comparison

**Part-V**

1) ~~imp~~install lazypredict (lazyregressor)
   ↳ Library
2) Load dataset
3) Data preprocessing
4) Compare ML algorithms (80% train 20% test)
5) Data visualization & model Performance
6) Calculation time

Part-VI

## Model Deployement

Activate conda → folder → cd streamlit

↓

cd bioactpredapp

↓

streamlit run app.py

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29/01/2026

After running colab

| Rank | Model | $R^2$ |
|---|---|---|
| 1 | XGBoost | 0.832 |
| 2 | RandomForest | 0.831 |
| 3 | LightGBM | 0.828 |
| 4 | CatBoost | 0.806 |
| 5 | Ridge | 0.430 |

Selected model for predicting

## Next steps :-

choose & build app with streamlit & check if model is predicting $IC_{50}$ / $PIC_{50}$

$IC_{50}$ (or) $pIC_{50}$ check

$$pIC_{50} = -\log_{10}(IC_{50})$$

4 — 10

Range

10 μm to 10 nm

Range

The model is predicting $IC_{50}$.

Converted $pIC_{50}$ values: [7.6.301303]